

(Mis)using the Lattice Isomorphism Problem

Cryptanalysis of the double-LIP and Construction of LIP-Based Blind Signatures

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Abstract. We explore the design of blind schemes based on the Lattice Isomorphism Problem (LIP), a recently proposed group-action-based assumption for post-quantum cryptography. Our work highlights both the potential and the limitations of LIP-based constructions. In particular, we analyze the Abe–Okamoto framework and demonstrate that it does not yield a secure instantiation under LIP. We further present an attack on the double LIP problem, allowing from two generic distinct instances of a LIP problem to recover the secret unimodular matrix. Finally, we propose a new blind signature scheme that combines LIP with the Closest Vector Problem (CVP) and a modular version of the Short Integer Solution (SIS) problem, offering a fresh direction for lattice-based blind constructions.

Keywords: Lattice isomorphism problem, cryptanalysis, blind signatures

1 Introduction

Lattice Isomorphism Problem has been studied in [DvW22] for its applications in cryptography. The authors’ principal motivation was to extend the principles of the McEliece cryptosystem to the lattice setting. Specifically, in McEliece they define a lattice with strong decoding capabilities as the private key, while the public key is derived through specific manipulations of this lattice.

Applications of LIP to Advanced Signatures. Blind signatures were introduced by Chaum [Cha82] and allow a user to obtain a signature on a message from a signer without the signer learning anything about the content of the message. The initial use of blind signatures was motivated by the e-cash systems. Today, blind signatures have more applications in privacy-preserving technologies, including digital currency, e-voting protocols [RAG07] and anonymous credentials [CHL05, Bra93, BL13, BCC⁺09]. A blind signature scheme is an interactive protocol between a signer S holding a secret (signing) key sk and a user U holding a public key pk and a message m . It satisfies three key properties: correctness, unforgeability, and blindness. The correctness property ensures that a valid signature σ can be verified using the public key pk and the message m . The unforgeability property requires that an adversary who makes k signing

queries can produce at most k valid signatures. Finally, the blindness property ensures that the signer cannot link a signature σ to any specific execution of the protocol.

Numerous blind signature schemes have been proposed based on various assumptions, but achieving efficient constructions remains a challenging task. Most blind signature schemes follow either the Pointcheval and Stern framework [PS96], based on Schnorr’s signatures [Sch91], or the Abe-Okamoto framework [Oka92], based on the Okamoto-Schnorr signature.

Post-Quantum Blind signatures There has been an extensive research on blind signatures from quantum-safe assumptions. In [KLLQ23] the authors presented the first isogeny-based blind signature scheme which is based in the Abe-Okamoto [Oka92] framework. This construction uses the property of a quadratic twist of an elliptic curve in such a way that one obtains isogenies with a richer structure than abstract group actions. A similar blind signature framework was used in the construction of the first code-based blind signature scheme in [KLP25]. The scheme is based on the Matrix Equivalence Digital Signature (MEDS) group action problem. Although the construction uses similar framework as the isogeny-based scheme in [KLLQ23], the code-based construction requires additional public key and commitment information to address the challenges that are specific to the MEDS group action. These challenges include the absence of a module structure (unlike Schnorr), the lack of a quadratic twist (unlike CSI-OTter), and the non-commutativity of the acting group. The first lattice-based blind signature scheme was proposed by Rückert [Rüc10], building on Lyubashevsky’s identification scheme [Lyu09]. However, this scheme was later shown to have significant security flaws. In 2022, Pino and Katsumata introduced a more recent scheme [dPK22], which is based on standard lattice assumptions. The primary drawback of this construction is its relatively large signature size of 100 KB. In the same year, a new construction [AKSY22] was proposed which is based on the one-more-ISIS assumption and offers a shorter signature size of only 45KB. Recently, a new approach of a lattice-based blind signature scheme was proposed in [JS25]. The last work follows a different design but the final construction achieves improvements in the signature size and computational complexity.

Our contribution While the two aforementioned constructions in the isogeny-based and code-based settings rely on cryptographic group actions, it would be natural to explore a blind signature construction within the framework of the lattice isomorphism problem (LIP). We first study the challenges and open problems of an Abe-Okamoto-type blind signature scheme whose security is based on LIP. The main building block of this construction is an OR proof. The OR proof is derived from the standard Schnorr-like Zero-Knowledge Proof (ZKP). In the LIP setting, the Schnorr protocol has been presented in [DvW22] as follows. The secret statement of the proof consists of a unimodular matrix U , while the public statement is given by two quadratic forms $Q_0, Q_1 \in \mathcal{S}_n^{>0}(\mathbb{Z})$, such that $Q_1 = U^t Q_0 U$ holds. The main goal of the proof is to prove the knowledge of U given public Q_0, Q_1 . The ZKP follows the basic Schnorr’s framework, where

in the first stage of the proof the prover commits to a randomly chosen secret matrix V by running a special sampling algorithm described in [DvW22] and computing $Q' = V^t Q_0 V$. Then, after receiving a challenge $c \in \mathcal{U}(\{0, 1\})$, the prover computes a response (the proof) as $W = U^{-c} V$. To verify the proof, one checks the equality $Q' = W^t Q_c W$. The above-mentioned OR proof can be directly derived from the standard Schnorr-like ZKP, where the prover has to convince the verifier of knowledge of one of the secret keys corresponding to one of the two given public keys. The main difference is the number of public keys, which are $Q^{(1)} = U^t \bar{Q} U$ and $Q^{(-1)} = U^{-1} \bar{Q} U^{-t}$ and the corresponding verification equations which are $\tilde{Q}^{(1)} = \tilde{W}^t Q^{(c)} \tilde{W}$ and $\tilde{Q}^{(-1)} = \tilde{W}^{-1} Q^{(-c)} \tilde{W}^{-t}$. We demonstrate that although the OR protocol closely resembles a Schnorr-like ZKP, the construction is not straightforward. For the verification equations to hold, the matrices U must be symmetric. However, since the secrets are sampled using a specific algorithm, this algorithm cannot impose any structural constraints on the sampled matrices.

Furthermore, we present an attack on the LIP problem that uses two instances (Q_1, Q_2) and $(\bar{Q}_1, \bar{Q}_2)$ to recover the secret key U . A related but weaker attack was independently discovered in [BCDF25]. In particular, [BCDF25] does not address whether the secret can be recovered from only two instances, leaving this as an open question. In contrast, we show that such recovery is indeed possible.

The attack considers a double instance of the LIP problem satisfying

$$\bar{Q}_1 = U^t Q_1 U \quad \text{and} \quad \bar{Q}_2 = U^t Q_2 U,$$

for a secret matrix U . The key observation is that the matrices

$$A = Q_2^{-1} Q_1 \quad \text{and} \quad \bar{A} = \bar{Q}_2^{-1} \bar{Q}_1$$

are similar, since $\bar{A} = U^{-1} A U$. Exploiting this similarity, we show how to recover a valid unimodular matrix U .

Unlike [BCDF25], which only demonstrates that LIP is not 3-weakly unpredictable, our method recovers the secret from two instances and therefore shows that LIP is not 2-weakly unpredictable. The proof relies on the fact that the matrix

$$A = Q_2^{-1} Q_1$$

always has real eigenvalues, and hence admits a real eigenbasis. When the eigenvalues are simple, we show how to recover all secret unimodular matrices U .

In particular, we prove that the number of such matrices is exactly 2^k , where k is the number of connected components of a graph on n vertices (the dimension of A).

Finally, since the blind signature from LIP cannot be realized within Abe-Okamoto's framework in the current setting, we consider an alternative approach. We recall the well-known RSA blind signature construction from [BNPS01]. The protocol is initiated by the user who on input a message m first blinds it by computing $H(m)$ and then rerandomizes it as $H(m)r^e := \tilde{M}$ for a randomness r and RSA public key and a collision-resistant hash function $H(\cdot)$. This

“blinded” message is forwarded to the signer who creates a signature by computing $\bar{M}^d =: \sigma_{\text{bl}}$, where d is the RSA secret key. The signer sends this message back to the user who can unblind it by computing $r^{-1}\sigma_{\text{bl}} := \sigma$. This σ is then the output signature of this protocol. We note, that the signer signed the blinded message and the blindness is based on the hardness of the RSA problem. The signature is unforgeable as shown in [BNPS01].

Motivated by this simple idea we present an adopted version of the RSA blind signature scheme to the LIP setting. We use the Hawk signature scheme [DvW22] as the underlying construction. In Hawk scheme the signature is constructed by first hashing the message into $h := H(m)$ and then computing $c = Uh$, where U is the unimodular matrix from $GL_n(\mathbb{Z})$ which is used as the secret key. Then, the signer samples $\sigma' \leftarrow \mathcal{D}_{S,\rho,c}$ according to the sampling algorithm specified in [DvW22] and computes the signature as $\sigma = U^{-1}\sigma'$. The verification procedure computes the P -norm of the vector $h - \sigma$, and checks that it is smaller than a given parameter ρ , i.e. $\|h - \sigma\|_P \leq \rho$, where $P = U^T S U$ is a public key for a public parameter S .

For our blind signature construction we start by hashing the message into $h := H(m)$ and then split this h into ℓ parts $h_1, \dots, h_\ell$, such that $h = \sum_{i=1}^{\ell} a_i h_i$. The coefficients a_i - small and mostly zero - are kept secret by the user and only the vector $(h_1, \dots, h_\ell)$ is sent to the signer. The signer runs the Hawk signing procedure on each component h_i and outputs ℓ blinded signatures $\sigma_1, \dots, \sigma_\ell$. The user unblinds the signature by computing a linear combination of σ_i with the same coefficients a_i and verifies it using Hawk’s verification procedure. Finally, we provide a full proof of security.

2 Preliminaries

In this section, we first review the main definitions related to the lattice isomorphism problem. Next, we recall the definitions of the two advanced signature schemes that we will focus on in this work.

2.1 Lattice Isomorphism Problem

A n -dimensional lattice Λ is a discrete additive subgroup of $\mathbb{R}^n$. For an integer $m \leq n$ and a basis matrix $\mathbf{B} \in \mathbb{R}^{n \times m}$ of rank m , $\Lambda(\mathbf{B}) = \{\mathbf{B}\mathbf{x} \in \mathbb{R}^n | \mathbf{x} \in \mathbb{Z}^m\}$ is the lattice generated by the column vectors (i.e. basis vectors) of $\mathbf{B}$. If $n = m$, the lattice $\Lambda(\mathbf{B})$ is called full-rank. The set of lattices can be thought of as the space $GL_n(\mathbb{R})/GL_n(\mathbb{Z})$.

Two lattices are isomorphic if there exists an orthonormal transformation $O \in \mathcal{O}_n(\mathbb{R})$ that sends one lattice to another.

Definition 1 (Lattice Isomorphism Problem (LIP)). *Given two isomorphic lattices $\mathcal{L}, \mathcal{L}' \subset \mathbb{R}^n$ find an orthonormal transformation $O \in \mathcal{O}_n(\mathbb{R})$ such that $\mathcal{L}' = O \cdot \mathcal{L}$.*

Quadratic Forms: A quadratic form is a positive definite real symmetric matrix. Consider a Gram map $\gamma : GL_n(\mathbb{R}) \rightarrow \mathcal{S}_n^{>0}(\mathbb{R})$; $B \mapsto Q = B^t B$ for a basis matrix B . The preimages of $\gamma(B)$ are exactly OB for $O \in \mathcal{O}_n(\mathbb{R})$. The inner product is given by $\langle \mathbf{x}, \mathbf{y} \rangle_Q := \mathbf{x}^t Q \mathbf{y}$ and the norm by $\|\mathbf{x}\|_Q^2 := \mathbf{x}^t Q \mathbf{x}$. The first lattice minimum is denoted by $\lambda_1(Q) := \min_{\mathbf{x} \in \mathbb{Z}^n \setminus 0} \|\mathbf{x}\|_Q$.

Let $\mathcal{S}_n^{>0}(\mathbb{R})$ be the realization of the quotient $\mathcal{O}_n(\mathbb{R}) \setminus GL_n(\mathbb{R})$. Then, the action of $U \in GL_n(\mathbb{Z})$ is given by $Q \mapsto U^t Q U$. With this information the LIP problem in Definition 1 can be rephrased as follows: If B, B' are two lattice bases, and let $B' = OBU$, then for $Q := B^t B$ it holds that $Q' := (B')^t B' = U^t B^t O^t O B U = U^t Q U$. We say that Q and Q' are equivalent if there exists such a unimodular matrix $U \in GL_n(\mathbb{Z})$ which satisfies the last equation. The equivalence class is denoted by $[Q]$.

Definition 2 (Decisional LIP). Let $\mathcal{L}_0$ and $\mathcal{L}_1$ be non-isomorphic lattices, with the corresponding quadratic forms Q_0 and Q_1 . Given a quadratic form $Q \in [Q_b]$, where $b \in \{0, 1\}$ is a uniformly random bit, the objective is to determine the bit b .

In the next definition, we recall the worst-case search LIP problem (wc-sLIP)

Definition 3 (wc-sLIP^Q [DvW22]). For a quadratic form $Q \in \mathcal{S}_n^{>0}$ the problem wc-sLIP^Q is, given any quadratic form $Q' \in [Q]$, to find a unimodular matrix $U \in GL_n(\mathbb{Z})$ such that $Q' = U^t Q U$.

The distinguishing version of the LIP problem is defined as follows:

Definition 4 (wc- Δ LIP^{Q₀, Q₁} [DvW22]). For two quadratic forms $Q_0, Q_1 \in \mathcal{S}_n^{>0}$ the problem wc- Δ LIP^{Q₀, Q₁} is, given any quadratic form $Q' \in [Q_b]$, $b \in \{0, 1\}$ to find b .

The average-case version of the lattice isomorphism problem is defined as follows

Definition 5 (ac-sLIP_s^Q [DvW22]). For a quadratic form $Q \in \mathcal{S}_n^{>0}$ and $s > 0$ the problem ac-sLIP_s^Q is, given a quadratic form sampled as $Q' \xleftarrow{s} \mathcal{D}_s([Q])$ (for a Gaussian form distribution $\mathcal{D}_s([Q])$ over $[Q]$), to find a unimodular $U \in GL_n(\mathbb{Z}_q)$ such that $Q' = U^t Q U$.

Definition 6 (ac- Δ LIP_s^{Q₀, Q₁} [DvW22]). For two quadratic forms $Q_0, Q_1 \in \mathcal{S}_n^{>0}$ and $s > 0$ the problem ac- Δ LIP_s^{Q₀, Q₁} is, given a quadratic form sampled as $Q' \xleftarrow{s} \mathcal{D}_s([Q_b])$, where $b \in \{0, 1\}$, to find b .

Definition 7 (Closest Vector Problem (CVP_γ)). The Closest Vector Problem is: given an n -dimensional lattice $\mathcal{L}$ and a vector $\mathbf{w} \in \text{span}(\mathcal{L})$ find a vector $\mathbf{v} \in \mathcal{L}$ such that

$$\|\mathbf{v} - \mathbf{w}\| \leq \gamma(n) \|\mathbf{u} - \mathbf{w}\|$$

for any other vector $\mathbf{u} \in \mathcal{L}$.

Definition 8 (Gap-CVP (GapCVP_γ)). The decision version of CVP_γ is to distinguish between triples $(\mathcal{L}, \mathbf{w}, d)$ for which there exists a lattice vector within distance d from $\mathbf{w}$ and triples for which there exists no lattice vector within distance d from $\mathbf{w}$.

Definition 9 (Smoothing parameter [DvW22]). For quadratic form $Q \in \mathcal{S}_n^{>0}$ and $\epsilon > 0$ we define the smoothing parameter $\eta_\epsilon(Q)$ as the minimal $s > 0$ such that $\rho_{Q^{-1}, 1/s}(\mathbb{Z}^n) \leq 1 + \epsilon$, where $\rho_{Q^{-1}, 1/s}$ denotes the Gaussian function with parameter $1/2$.

Lemma 1 (Smoothing bound [DvW22, MR04]). For any quadratic form $Q \in \mathcal{S}_n^{>0}$, we have $\eta_{2^{-1}}(Q) \leq \sqrt{n}/\lambda_1(Q^{-1})$ and $\eta_\epsilon(Q) \leq \|B_Q^*\| \cdot \sqrt{\ln(2n(1 + 1/\epsilon))}/\pi$ for $\epsilon > 0$, where $\rho_{Q, s, \mathbf{c}}(\mathbf{x}) = \exp(-\pi\|\mathbf{x} - \mathbf{c}\|_Q^2/s^2)$ is the Gaussian function on $\mathbb{R}^n$.

Lemma 2 (Tailbound [DvW22, MR04]). For any quadratic form $Q \in \mathcal{S}_n^{>0}$, $\epsilon \in (0, 1)$, center $\mathbf{c} \in \mathbb{R}^n$ and parameter $s \geq \eta_\epsilon(Q)$, we have $\Pr_{\mathbf{x} \sim \mathcal{D}_{Q, s, \mathbf{c}}}[\|\mathbf{x} - \mathbf{c}\|_Q > s\sqrt{n}] \leq \frac{1+\epsilon}{1-\epsilon} \cdot 2^{-n}$.

Algorithm 1 Sampling from $\mathcal{D}_s([Q])$

Input: A quadratic form $Q \in \mathcal{S}_n^{>0}(\mathbb{Z}^n)$, and a parameter
1: $s \geq \max\{\lambda_n(Q), \|B_Q^*\| \cdot \sqrt{\ln(2n + 4)}/\pi$.
Output: Sample $R = U^t Q U$ from $\mathcal{D}_s([Q])$, with a transformation $U \in GL_n(\mathbb{Z})$.
2: **for** $C \leftarrow 1 - (1 + e^{-\pi})^{-1}$, $m \leftarrow \lceil \frac{2n}{C} \rceil$ **do**
3: $\mathbf{y}_1, \dots, \mathbf{y}_m \leftarrow \mathcal{D}_{Q, s}$
4: $Y \leftarrow (\mathbf{y}_1, \dots, \mathbf{y}_m)$
5: **while** $\text{rk}(Y) < n$; **do**
6: $(R, U) \leftarrow \text{Extract}(Q, Y)$

Lemma 3 (Discrete Sampling [DvW22, BLP⁺13]). There is a polynomial-time algorithm $\text{DiscreteSample}(Q, s, \mathbf{c})$, that given a quadratic form $Q \in \mathcal{S}_n^{>0}$, center $\mathbf{c} \in \mathbb{R}^n$, and a parameter $s \geq \|B_Q^*\| \cdot \sqrt{\ln(2n + 4)}/\pi$, returns a sample distributed as $\mathcal{D}_{Q, s, \mathbf{c}}$.

Lemma 4 ([DvW22]). For any quadratic form $Q \in \mathcal{S}_n^{>0}$, parameter $\epsilon \in (0, 1)$, and $s \geq \max\{\lambda_n(Q), \eta_\epsilon(Q)\}$ we have

$$\Pr_{Q' \sim \mathcal{D}_s([Q])}[\|B_{Q'}^*\| > s\sqrt{n}] \leq \frac{1 + \epsilon}{1 - \epsilon} \cdot 100n \cdot 2^{-n}.$$

Definition 10 (Extractor [DvW22]). An efficient function $\mathcal{E} : \mathcal{X} \times \{0, 1\}^z \rightarrow \{0, 1\}^v$ is an (m, ϵ) -extractor, if, for all random variables X distributed over $\mathcal{X}$ and $H_\infty(X) \geq m$, it holds that $\Delta((Z, \epsilon(X, Z)), (Z, V)) \leq \epsilon$, where the seed $Z \leftarrow \mathcal{U}(\{0, 1\}^z)$ and $V \leftarrow \mathcal{U}(\{0, 1\}^v)$ are drawn uniformly at random, and independently of X .

2.2 Sigma Protocol

Definition 11 (Sigma Protocol). A sigma protocol for an NP relation $\mathcal{R} \subseteq \{0, 1\}^* \times \{0, 1\}^*$ is a three-move interactive protocol which consists of two PPT algorithms $P = (P_1, P_2)$ and V which works as follows:

1. On input $(X, W) \in \mathcal{R}$, the prover runs algorithm P_1 to obtain a commitment comm and state state . It sends comm to the verifier.
2. On input comm the verifier chooses a challenge $\text{chal} \in \mathcal{C}$ uniformly at random and sends it to P .
3. On input $(\text{chal}, \text{state})$, the prover runs algorithm P_2 to construct a response resp which it sends to the verifier.
4. The verifier runs $V(X, \text{comm}, \text{chal}, \text{resp})$ and outputs a bit b , indicating whether the proof is valid (1) or invalid (0).

A correct and secure sigma protocol satisfies the following properties:

Correctness: If the prover and verifier follow the protocol description, then the verifier will output 1 with probability 1.

Honest Verifier Zero-Knowledge (HVZK): There exists an efficient PPT simulator Sim that, given a statement X , outputs a valid transcript $(\text{comm}, \text{chal}, \text{resp})$, in such a way that the simulator's distribution of outputs is identical to the distribution of outputs of honest transcripts.

Special Soundness: There exists a deterministic polynomial-time extractor Ext that, given two valid transcripts $(\text{comm}, \text{chal}_0, \text{resp}_0)$ and $(\text{comm}, \text{chal}_1, \text{resp}_1)$ with respect to a statement X , satisfying $\text{chal}_0 \neq \text{chal}_1$, outputs a witness $\hat{W}$ such that $(X, \hat{W}) \in \mathcal{R}$.

A weaker notion of HVZK is witness indistinguishability, which requires that for any statement X in the language of $\mathcal{R}$ and any witnesses W_1, W_2 such that $(X, W_1), (X, W_2) \in \mathcal{R}$, the distribution of honest transcripts arising when P has (X, W_1) is identical to the distribution of honest transcripts when P has (X, W_2) .

Zero-Knowledge Proof (ZKP) of Knowledge Σ [DvW22]. The protocol 1 relies on a group acting $GL_n(\mathbb{Z})$ which does not admit uniform distribution. However, uniformity is not required in the following Zero-Knowledge Proof.

The public input to the ZKP consists of two equivalent quadratic forms $Q_0, Q_1 \in \mathcal{S}_n^{>0}(\mathbb{Z})$ and the secret input of the prover is a unimodular $U \in GL_n(\mathbb{Z})$, s.t. $Q_1 = U^t Q_0 U$. The ZKP is defined for the following parameter bound:

$$s \geq \max\{\lambda_n([Q_0]), \max\{\|B_{Q_0}^*\|, \|B_{Q_1}^*\| \cdot \sqrt{\ln(2n+4)/\pi}\}\}.$$

2.3 Blind Signature Scheme - Abe-Okamoto Framework

We recall a definition of the Abe-Okamoto framework blind signature scheme in a random oracle model. This scheme consists of four moves, starting with a

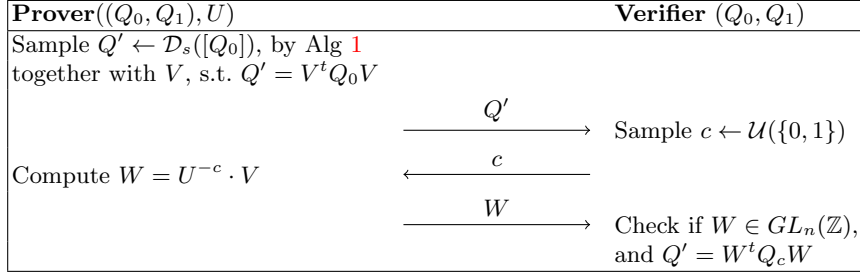


Fig. 1. Zero-Knowledge Proof of Knowledge Σ

signer committing to the secret key. This commitment is then sent to the user who blinds the message and returns it to the signer. The signer generates a signature on the blinded message and sends it to the user. The user unblinds the message and returns a signature on the original message. Upon receiving the signature, a verifier runs through the verification steps and outputs either 1 or 0.

Definition 12 (Blind Signature). A blind signature scheme consists of the four algorithms $\Sigma_{BS} = (\text{BS.KGen}, \text{BS.S}, \text{BS.U}, \text{BS.Verify})$ which are specified as follows:

- $\text{BS.KGen}(1^\lambda)$: On input a security parameter 1^λ it outputs a private signing key sk and a public verification key pk
- $\text{BS.S} = (\text{BS.S}_1, \text{BS.S}_2)$: Is an interactive protocol consisting of two phases.
 - $\text{BS.S}_1(\text{sk})$: On input a secret key sk , it outputs an internal state state_S and a first-sender response $\rho_{S,1}$.
 - $\text{BS.S}_2(\text{state}_S, \rho_U)$: On input a state state_S , and a user message ρ_U , it outputs a second-sender response $\rho_{S,2}$.
- $\text{BS.U} = (\text{BS.U}_1, \text{BS.U}_2)$: is an interactive protocol consisting of two phases:
 - $\text{BS.U}_1(\text{pk}, \text{msg}, \rho_{S,1})$: On input a public key pk , a message msg , and a response $\rho_{S,1}$, it outputs an internal user state state_U and a user message ρ_U .
 - $\text{BS.U}_2(\text{state}_U, \rho_{S,2})$: On input a user state state_U and a second-sender response $\rho_{S,2}$ it outputs a signature σ on a message msg .
- $\text{BS.Verify}(\text{pk}, \text{msg}, \sigma)$: On input a public key pk , a message msg and a signature σ , it outputs 1 if σ is a valid signature on msg or $\perp$ otherwise.

Completeness: A blind signature scheme Σ_{BS} is complete if for any $\lambda \in \mathbb{N}$ and $\text{msg} \in \mathcal{M}$ the verification algorithm $\text{BS.Verify}(\text{pk}, \text{msg}, \sigma) = 1$ with overwhelming probability if $(\text{sk}, \text{pk}) \leftarrow \text{BS.KGen}(1^\lambda)$, $(\perp, \sigma) \leftarrow \text{BS.S}(\text{msg}, \text{sk})$.

Blindness: For a four-move blind signature scheme Σ_{BS} , we define the blindness game Blind_{BS} with an adversary $\mathcal{S}$ (playing the signer) as follows:

Setup: Sample a bit $b \xleftarrow{\$} \{0, 1\}$. Then run $\mathcal{S}$ on input 1^λ .

Online Phase: When $\mathcal{S}$ outputs messages $\tilde{\text{msg}}_0, \tilde{\text{msg}}_1$ and a public key pk , the game checks if pk is valid and if so, it assigns $\text{msg}_0 := \tilde{\text{msg}}_b, \text{msg}_1 := \tilde{\text{msg}}_{1-b}$. If pk is not valid, the game aborts and outputs 0. $\mathcal{S}$ is given access to oracles $\mathcal{O}_{U_1}$ and $\mathcal{O}_{U_2}$, which work as follows:

Oracle $\mathcal{O}_{U_1}$: On input a bit $b' \in \{0, 1\}$ and a S_1 response $\rho_{S,1}$, if the session b' is not yet open, the oracle marks session b' as open and generates a state and a challenge as $(\text{state}_{U,b'}, \rho_{U,b}) \xleftarrow{\$} \text{BS.U}_1(\text{pk}, \text{msg}_{b'}, \rho_{S,1})$. It returns $\rho_{U,b}$ to $\mathcal{S}$. Otherwise, it returns $\perp$.

Oracle $\mathcal{O}_{U_2}$: On input a bit $b' \in \{0, 1\}$ and a second-signer message $\rho_{S,2,b'}$, if the session b' is open, it computes a signature $\sigma_{b'} := \text{BS.U}_2(\text{pk}, \text{state}_{U,b'}, \rho_{S,2,b'})$. It marks session b' as closed and saves $\sigma_{b'}$. If both sessions are closed and they have produced signatures, the oracle outputs the two signatures σ_0, σ_1 to $\mathcal{S}$.

Output Determination. If both sessions are closed and signatures are produced, the game outputs 1 if $\mathcal{S}$ outputs a bit $b' = b$. Otherwise, return 0.

We define the advantage of $\mathcal{S}$ as $\text{Adv}_{\mathcal{S}, \text{BS}}^{\text{Blind}} = |\Pr[\text{Blind}_{\text{BS}} = 1] - 1/2|$, where the probability goes over the randomness of the game as well as the randomness of the adversary $\mathcal{S}$. We say the scheme Σ_{BS} is (t, ϵ) - blind if for any adversary $\mathcal{S}$ running in time t ,

$$\text{Adv}_{\mathcal{S}, \text{BS}}^{\text{Blind}} \leq \epsilon.$$

One-More Unforgeability: Let $\Sigma_{\text{BS}} = (\text{BS.KGen}, \text{BS.S}, \text{BS.U}, \text{BS.Verify})$ be a blind signature and λ be the security parameter. We define the ℓ -one more unforgeability game $\ell - \text{OMUF}_{\text{PBS}}$ with an adversary $\mathcal{A}$ (playing the user) as follows:

Setup: Sample a pair of keys $(\text{sk}, \text{pk}) \xleftarrow{\$} \text{BS.KGen}(1^\lambda)$. Initialize $\ell_{\text{closed}} := 0$ and run $\mathcal{A}$ on input pk .

Online Phase: $\mathcal{A}$ has access to oracles $\mathcal{O}_{S_1}$ and $\mathcal{O}_{S_2}$, which work as follows:

Oracle $\mathcal{O}_{S_1}$: The oracle samples a fresh session identifier sid . It sets $\text{open}_{\text{sid}} := \text{true}$ and generates $(\text{state}_{S,\text{sid}}, \rho_{S,1}) \xleftarrow{\$} \text{BS.S}_1(\text{sk})$. Then it returns the response $\rho_{S,1}$ to $\mathcal{A}$.

Oracle $\mathcal{O}_{S_2}$: If $\ell_{\text{closed}} < \ell$ the oracle takes as input a user message ρ_U and a session identifier sid . If $\ell_{\text{closed}} \geq \ell$ or $\text{open}_{\text{sid}} = \text{false}$, it returns $\perp$. Otherwise, it sets $\ell_{\text{closed}}++$ and $\text{open}_{\text{sid}} := \text{false}$. Then it computes the response $\rho_{S,2} \xleftarrow{\$} \text{BS.S}_2(\text{state}_{S,\text{sid}}, \rho_U)$ and returns $\rho_{S,2}$ to $\mathcal{A}$.

Output Determination. When $\mathcal{A}$ outputs distinct tuples $(\text{msg}_1, \sigma_1), \dots, (\text{msg}_k, \sigma_k)$, return 1 if $k \geq \ell_{\text{closed}} + 1$ and $\text{BS.Verify}(\text{pk}, \text{msg}_i, \sigma_i) = 1$ for all $1 \leq i \leq k$. Otherwise, return 0.

We define the advantage of $\mathcal{A}$ as $\text{Adv}_{\mathcal{A}, \text{BS}}^{\ell - \text{OMUF}_{\text{BS}}}(\lambda) = \Pr[\ell - \text{OMUF}_{\text{BS}} = 1]$, where the probability goes over the randomness of the game as well as the randomness of the adversary $\mathcal{A}$. We say the scheme Σ_{BS} is ℓ -one-more unforgeable if for any adversary $\mathcal{A}$ that makes at most ℓ queries to $\mathcal{O}_{S_1}$ the following holds

$$\text{Adv}_{\mathcal{A}, \text{BS}}^{\ell - \text{OMUF}_{\text{PBS}}}(\lambda) \leq \text{negl}(\lambda).$$

2.4 RSA Blind Signature Scheme

Definition 13 (Blind Signature [BNPS01]). A blind signature scheme consists of the following three algorithms $\Sigma_{\text{RSA-BS}} = (\text{RSA-BS.KGen}, \text{RSA-BS.Sign}, \text{RSA-BS.Verify})$:

- $\text{RSA-BS.KGen}(1^\lambda)$: On input a security parameter 1^λ the algorithm outputs RSA secret and public key, $\text{sk} = (d, p, q)$ and $\text{pk} = (e, n)$ where $n = pq$ with p, q prime.
- $\text{RSA-BS.Sign}(m, \text{sk})$: It is an interactive protocol between a user and a signer. The user chooses a random $r \xleftarrow{\$} \mathbb{Z}_n^*$, s.t. $\gcd(r, n) = 1$ and computes $m' = H(m)r^e$, where $H : \{0, 1\} \rightarrow \mathbb{Z}_n$ is a hash function. The user sends m' to the signer. Using the secret key d , the signer computes $s' = (m')^d \bmod n$ and sends it back to the user. The user unblinds the signature by computing $s = s'r^{-1} \bmod n$.
- $\text{RSA-BS.Verify}(\text{pk}, s, m)$: The verifier uses the signer's public key (e, n) to verify the signature by computing $s^e \bmod n = H(m)$.

We will provide an adapted version of this RSA based blind signature scheme to the LIP setting.

Security Definitions of RSA Blind Signature.

Definition 14 (Blindness). For any probabilistic polynomial-time adversary $\mathcal{A}$, the probability that the following blindness experiment evaluates to 1 (i.e. the adversary succeeds in breaking the blindness property) is

$$\text{Adv}_{\Sigma_{bl}}^{\text{blind}}(\mathcal{A}) := \Pr[\text{Blind}_{\Sigma_{bl}}(\mathcal{A}, \lambda) = 1] \leq \frac{1}{2} + \text{negl}(\lambda).$$

Experiment $\text{Blind}_{\Sigma_{bl}}(\mathcal{A}, \lambda)$

$\text{pk}, \text{sk} \leftarrow \text{BS.KGen}(1^\lambda)$
 $(m_0, m_1, \beta_{\text{find}}) \leftarrow \mathcal{A}(\text{find}, \text{pk})$
 $b \leftarrow \{0, 1\}$
 $\beta_{\text{issue}} \leftarrow \mathcal{A}^{\text{BS.Sign}(\text{sk}, \cdot)}(\text{issue}, \beta_{\text{find}})$
 let σ_b, σ_{1-b} be the local outputs of BS.Sign on input m_b and m_{1-b} respectively.
 $b' \leftarrow \mathcal{A}(\text{guess}, \sigma_0, \sigma_1, \beta_{\text{issue}})$
 Return a random bit if $\sigma_0 = \perp$ or $\sigma_1 = \perp$, else return 1 iff $b = b'$

Definition 15 (Unforgeability). For any probabilistic polynomial-time adversary $\mathcal{A}$ the probability that experiment $\text{Forge}_{\Sigma_{bl}}$ evaluates to 1 is

$$\text{Adv}_{\Sigma_{bl}}^{\text{UNF}}(\mathcal{A}) := \Pr[\text{Forge}_{\Sigma_{bl}}(\mathcal{A}, \lambda) = 1] \leq \text{negl}(\lambda).$$

Experiment $\text{Forge}_{\Sigma_{bl}}(\mathcal{A}, \lambda)$

$(\text{sk}, \text{pk}) \leftarrow \text{KeyGen}(1^\lambda)$
 $((m_1, \sigma_1), \dots, (m_{k+1}, \sigma_{k+1})) \leftarrow \mathcal{A}^{\text{BS.Sign}(\text{sk}, \cdot)}(\text{pk})$
 Return 1 iff
 $m_i \neq m_j$ for $1 \leq i < j \leq k+1$, and
 $\text{Verify}(\sigma_i, m_i, \text{pk}) = 1$ for all $i = 1, 2, \dots, k+1$ and
 at most k interactions with $\text{BS.Sign}(\text{sk}, \cdot)$ were initiated.

3 LIP-Based Blind Signature Scheme

In this section, we present an attempt of constructing a blind signature scheme based on the Lattice Isomorphism Problem (LIP). The scheme follows the well-known Abe–Okamoto framework. We present the challenges and show that constructing a blind signature scheme in this framework is not feasible due to the inherent asymmetry of the secret key.

An Abe–Okamoto-style blind signature scheme fundamentally relies on an OR-proof as its core building block. We introduce our approach to designing such a scheme based on LIP; however, we demonstrate that this construction is infeasible in the current setting. The key limitation lies in the sampling algorithm described in Section 1, which does not generate symmetric matrices, a requirement to achieve the necessary cryptographic properties. This observation highlights an important structural requirement of such constructions within this framework and explains why the straightforward approach does not work with the present sampling method.

Furthermore, we present a cryptanalysis of a blind signature scheme based on LIP and conclude that the Abe–Okamoto framework cannot be instantiated under LIP-based assumptions.

3.1 Blind Signature in Abe–Okamoto Framework

We begin with the key components of the blind signature scheme: a Sigma protocol and its OR-composition.

Sigma protocol and OR-composition The OR-composition of a Sigma protocol allows a prover, given two public keys, to prove knowledge of the secret associated with one of them, while keeping the identity of the known secret hidden.

Sigma protocol: We first provide a sigma protocol for the following relation

$$\mathcal{R}_{\bar{Q}}^{\pm 1} = \left\{ \left(\begin{array}{l} X = (Q^{(1)}, Q^{(-1)}), \\ W = U \end{array} \right) : \begin{array}{l} Q^{(1)} = U^t \bar{Q} U, \\ Q^{(-1)} = U^{-1} \bar{Q} U^{-t} \end{array} \right\},$$

where $Q^{(1)}, Q^{(-1)}$ and $\bar{Q}$ are in $\mathcal{S}_n^{>0}(\mathbb{Z})$ and U is a unimodular matrix from $GL_n(\mathbb{Z})$. Furthermore, a public parameter is defined as

$$s \geq \max\{\lambda_n([\bar{Q}]), \|B_{\bar{Q}}^*\| \cdot \sqrt{\ln(2n+4)/\pi}\},$$

Proposition 1. *The protocol of 2 is correct.*

Prover $((Q^{(1)}, Q^{(-1)}), U)$	Verifier $(Q^{(1)}, Q^{(-1)})$
Sample $\tilde{Q}^{(1)}, \tilde{Q}^{(-1)} \leftarrow \mathcal{D}_s([\tilde{Q}])$ by Alg 1 together with V s.t. $\tilde{Q}^{(1)} = V^t \tilde{Q} V$, and $\tilde{Q}^{(-1)} = V^{-1} \tilde{Q} V^{-t}$	
	$\xrightarrow{\tilde{Q}^{(1)}, \tilde{Q}^{(-1)}} \text{Sample } c \leftarrow \mathcal{U}(\{-1, 1\})$
Compute $\tilde{W} = U^{-c} \cdot V$	$\xleftarrow{c}$
	$\xrightarrow{\tilde{W}} \text{Check if } \tilde{W} \in GL_n(\mathbb{Z}),$ and $\tilde{Q}^{(1)} = \tilde{W}^t Q^{(c)} \tilde{W}$ $\tilde{Q}^{(-1)} = \tilde{W}^{-1} Q^{(-c)} \tilde{W}^{-t}$

Fig. 2. Our sigma protocol for the relation $\mathcal{R}_{\tilde{Q}}^{\pm}$

Proof. Because of the condition

$$s \geq \max\{\lambda_n([\tilde{Q}]), \|B_{\tilde{Q}}^*\| \cdot \sqrt{\ln(2n+4)/\pi}\},$$

the sampling algorithm runs in polynomial time. We show that as long as both parties follow the protocol description, the verification equations in protocol 2 are correct. Set $Q^{(1)} = U^t \tilde{Q} U$ and $Q^{(-1)} = U^{-1} \tilde{Q} U^{-t}$ and $\tilde{W} = U^{-c} V$.

For $c = 1$ check:

$$\begin{aligned} \tilde{Q}^{(1)} &= \tilde{W}^t Q^{(1)} \tilde{W} \\ V^t \tilde{Q} V &= V^t U^{-t} U^t \tilde{Q} U U^{-1} V = V^t \tilde{Q} V = \tilde{Q}^{(1)}, \end{aligned} \tag{1}$$

and

$$\begin{aligned} \tilde{Q}^{(-1)} &= \tilde{W}^{-1} Q^{(-1)} \tilde{W}^{-t} \\ V^{-1} \tilde{Q} V^{-t} &= V^{-1} U U^{-1} \tilde{Q} U^{-t} U^t V^{-t} = V^{-1} \tilde{Q} V^{-t} = \tilde{Q}^{(-1)}. \end{aligned} \tag{2}$$

Similarly we check the equations for $c = -1$.

$$\begin{aligned} \tilde{Q}^{(1)} &= \tilde{W}^t Q^{(-1)} \tilde{W} \\ V^t \tilde{Q} V &= V^t U^t U^{-1} \tilde{Q} U^{-t} U V = V^t \tilde{Q} V = \tilde{Q}^{(1)}, \end{aligned} \tag{3}$$

and

$$\begin{aligned} \tilde{Q}^{(-1)} &= \tilde{W}^{-1} Q^{(1)} \tilde{W}^{-t} \\ V^{-1} \tilde{Q} V^{-t} &= V^{-1} U^{-1} U^t \tilde{Q} U U^{-t} V^{-t} = V^{-t} \tilde{Q} V^{-1} = \tilde{Q}^{(-1)}. \end{aligned} \tag{4}$$

Here we note that the equations 3 and 4 are correct if and only if $U^t = U$.

Theorem 1. *The protocol in Figure 2 is 2-special sound (meaning that the witness is recoverable from 2 transcripts and 2 challenges, while the commitment remains the same)*

Proof. Set $\text{comm} = (\tilde{Q}^{(1)}, \tilde{Q}^{(-1)})$. Let $\tau = (\text{comm}, c, \tilde{W})$ and $\hat{\tau} = (\text{comm}, \hat{c}, \hat{W})$ be accepting transcripts. Then holds:

$$\begin{aligned}\tilde{Q}^{(1)} &= \tilde{W}^t Q^{(c)} \tilde{W} = \hat{W}^t Q^{(\hat{c})} \hat{W} \leftrightarrow \hat{W}^{-t} \tilde{W}^t Q^{(c)} = Q^{(\hat{c})} \hat{W} \tilde{W}^{-1} \\ \text{wlog } c = 1, \hat{c} = -1 \text{ yields:} \\ \hat{W}^{-t} \tilde{W}^t Q^{(1)} &= Q^{(-1)} \hat{W} \tilde{W}^{-1} \leftrightarrow Q^{(1)} = \tilde{W}^{-t} \hat{W}^t Q^{(-1)} \hat{W} \tilde{W}^{-1}.\end{aligned}$$

Similarly we check that the equations are satisfied for $c = -1$:

$$\begin{aligned}\tilde{Q}^{(-1)} &= \tilde{W}^{-1} Q^{(c)} \tilde{W}^{-t} = \hat{W}^{-1} Q^{(\hat{c})} \hat{W}^{-t} \leftrightarrow \hat{W} \tilde{W}^{-1} Q^{(c)} = Q^{(\hat{c})} \hat{W}^{-t} \tilde{W}^t \\ \text{wlog } c = 1, \hat{c} = -1 \text{ yields:} \\ \hat{W} \tilde{W}^{-1} Q^{(1)} &= Q^{(-1)} \hat{W}^{-t} \tilde{W}^t \leftrightarrow Q^{(1)} = \tilde{W} \hat{W}^{-1} Q^{(-1)} \hat{W}^{-t} \tilde{W}^t.\end{aligned}$$

So we see that $\tilde{W}^{-t} \hat{W}^t$ is a witness to $\mathbf{X} = (Q^{(1)}, Q^{(-1)})$.

A proposed OR-composition of the Sigma protocol is presented in Figure 3. The security of this composition relies on the security of the Sigma protocol shown in Figure 2. In the following paragraph, we explain why a secure instantiation of the Sigma protocol is not possible in the LIP setting.

Prover $((Q_b^{(1)}, Q_b^{(-1)}), U_b)$	Verifier $(Q_b^{(1)}, Q_b^{(-1)})$
Sample $\tilde{Q}_b^{(1)}, \tilde{Q}_b^{(-1)} \leftarrow \mathcal{D}_s([\tilde{Q}])$ by Alg. 1 together with V_b s.t. $\tilde{Q}_b^{(1)} = V_b^t \tilde{Q} V_b$, and $\tilde{Q}_b^{(-1)} = V_b^{-1} \tilde{Q} V_b^{-t}$ $c_{1-b} \xleftarrow{\$} \{-1, 1\}$ Sample $\tilde{Q}_{1-b} \leftarrow \mathcal{D}_s([Q_{1-b}])$, by Alg 1 together with R_{1-b} s.t. $\tilde{Q}_{1-b}^{(c_{1-b})} = R_{1-b}^t Q_{1-b} R_{1-b}$	
	$\xrightarrow{\tilde{Q}_0^{(1)}, \tilde{Q}_0^{(-1)}, \tilde{Q}_1^{(1)}, \tilde{Q}_1^{(-1)}} \text{Sample } c \leftarrow \mathcal{U}(\{-1, 1\})$
Compute $c_b = c \cdot c_{1-b}$ Compute $\tilde{R}_b = U_b^{-c} \cdot V$	
	$\xleftarrow{c}$ $\xrightarrow{\tilde{R}_0, \tilde{R}_1, c_0, c_1}$
	Check if: $c = c_0 \cdot c_1$ $\tilde{R}_b \in GL_n(\mathbb{Z})$, and $\tilde{Q}_b^{(1)} = \tilde{R}_b^t Q_b^{(c_b)} \tilde{R}_b$ $\tilde{Q}_b^{(-1)} = \tilde{R}_b^{-t} Q_b^{(-c_b)} \tilde{R}_b^{-1}$

Fig. 3. OR-Composition of our sigma protocol

On sampling symmetric key using Algorithm 1.

We observed that, in order for the verification equations to be satisfied, the secret key matrix U must be symmetric; that is, it must hold that $U = U^t$. This symmetry is crucial for ensuring the correctness of the protocol, particularly in the context of the transformations involved in the zero-knowledge proofs. However, the sampling algorithm described in Algorithm 1 does not impose any structural constraints on the sampled matrix U , and in particular, it does not ensure symmetry.

In Figure 6, we propose a construction of a blind signature scheme based on the Lattice Isomorphism Problem (LIP). However, we note that for this scheme to function correctly, the sampling algorithm must be adapted to generate symmetric matrices. Alternatively, one could explore other techniques or frameworks to circumvent this limitation, such as modifying the proof structure or introducing additional algebraic tools that relax the symmetry requirement while preserving security and correctness.

4 Cryptanalysis of the Double-LIP

In addition to the obstacle that the secret key needs to be a symmetric matrix as pointed out in the last section, we note that there exists an attack on the LIP-based Abe-Okamoto blind signature scheme which we describe in the next paragraphs. We note that a similar weakness was independently exposed in [BCDF25]. In it, the authors relate two instances of LIP with the same unimodular matrix U as in (5) (2-LIP) to the computation of the rank of the $2n^2 \times 2n^2$ matrix A given by

$$A = \begin{pmatrix} \bar{Q}_1 \otimes I_n & I_n \otimes (-Q_1) \\ \bar{Q}_2 \otimes I_n & I_n \otimes (-Q_2) \end{pmatrix}.$$

In particular, they show that if $\bar{Q}_1, \bar{Q}_2$ have no relation to Q_1, Q_2 , then $\text{rank } A = 2n^2$ with high probability, while in a 2-LIP instance this rank is always $\leq 2n^2 - n$. Our analysis is different, although it is based on the same observation that $Q_2^{-1}Q_1$ and $\bar{Q}_2^{-1}\bar{Q}_1$ are similar. In particular, in most cases, we recover a solution $U \in \text{GL}_n(\mathbb{Z})$. In the terminology of [BCDF25], our analysis shows that LIP is not 2-weakly unpredictable, improving their observation that LIP is not 3-weakly unpredictable.

Theorem 2 (2-weakly predictability of LIP). *Let Q_1, Q_2 be two given $n \times n$ real symmetric matrices, $U \in \text{GL}_n(\mathbb{Z})$ a secret unimodular matrix. Suppose that*

$$\bar{Q}_1 = U^t Q_1 U \quad \text{and} \quad \bar{Q}_2 = U^t Q_2 U \tag{5}$$

are given (2-LIP instance). Then it is in general¹ possible to recover all unimodular U satisfying (5).

¹ The general case is when the eigenvalues of $Q_2^{-1}Q_1$ are all distinct.

Proof. Note that

$$\bar{A} = \bar{Q}_2^{-1} \bar{Q}_1 = U^{-1} Q_2^{-1} Q_1 U = U^{-1} A U . \quad (6)$$

The matrix A , although not necessarily symmetric, is nevertheless diagonalizable over $\mathbb{R}$, since, calling $Q_1^{1/2}$ the square root matrix (a symmetric matrix) of the positive definite Q_1 , we have

$$Q_1^{1/2} A Q_1^{-1/2} = Q_1^{1/2} Q_2^{-1} Q_1 Q_1^{-1/2} = Q_1^{1/2} Q_2^{-1} Q_1^{1/2}$$

which is a real symmetric matrix, hence diagonalizable over the reals. In the generic case of distinct eigenvalues of A , we can construct a basis of eigenvectors $\{v_1, \dots, v_n\}$ for A , and similarly a basis $\{\bar{v}_1, \dots, \bar{v}_n\}$ for $\bar{A}$, relative to the same sequence of eigenvalues. Then (6) shows that there exist nonzero $c_1, \dots, c_n \in \mathbb{R}$ such that

$$U \bar{v}_i = c_i v_i , \quad i = 1, \dots, n ,$$

since, calling λ_i the eigenvalue of v_i and $\bar{v}_i$, we have

$$\lambda_i \bar{v}_i = \bar{A} \bar{v}_i = U^{-1} A U \bar{v}_i \Rightarrow \lambda_i U \bar{v}_i = A U \bar{v}_i ,$$

showing that $U \bar{v}_i$ is an eigenvector of A relative to λ_i , hence must be a multiple of v_i .

To find c_i , note that

$$\bar{v}_i^t \bar{Q}_1 \bar{v}_i = \bar{v}_i^t U^t Q_1 U \bar{v}_i = c_i^2 v_i^t Q_1 v_i \implies c_i^2 = \frac{\bar{v}_i^t \bar{Q}_1 \bar{v}_i}{v_i^t Q_1 v_i} .$$

This reveals c_i up to sign. Similarly,

$$\bar{v}_i^t \bar{Q}_1 \bar{v}_j = \bar{v}_i^t U^t Q_1 U \bar{v}_j = c_i c_j (v_i^t Q_1 v_j) . \quad (7)$$

Since U and $-U$ are both solutions of the 2-LIP instance, we can suppose that $\text{sign}(c_1) > 0$. The previous equation then says that

$$\text{sign}(c_j) = \text{sign} \left(\frac{\bar{v}_1^t \bar{Q}_1 \bar{v}_j}{v_1^t Q_1 v_j} \right) ,$$

provided $v_1^t Q_1 v_j \neq 0$. In general, (7) for Q_1 and Q_2 allows to determine the sign of c_j from the sign of c_i , provided, at least one of $v_i^t Q_1 v_j$ or $v_i^t Q_2 v_j$ is nonzero. This suggests looking at the graph with n vertices, labeled $c_1, \dots, c_n$, and edges between c_i and c_j if at least one of $v_i^t Q_1 v_j$ or $v_i^t Q_2 v_j$ is nonzero. If this graph is connected, then the previous discussion allows one to recover all the signs to get exactly the two solutions U and $-U$.

Otherwise, for each connected component of this graph, all signs of vertices c_j are uniquely determined from an arbitrary choice for one vertex c_i in the same component, without prejudice to the other connected components. Since all vectors v_i (resp. $\bar{v}_i$) relative to one connected component are orthogonal for both inner products Q_1 and Q_2 (resp. $\bar{Q}_1$ and $\bar{Q}_2$), we can split the double-LIP instance to k double-LIP instances, where k is the number of connected components, yielding exactly 2^k distinct solutions to the double-LIP instance.

5 A New Blind Signature Scheme

Our proposed blind signature scheme is based on the Hawk Signature scheme which was introduced in [DvW22]. We first recall this scheme in Figure 4 and in the following we provide our construction.

KGen(1^λ) :	sample $U \xleftarrow{\$} GL_n(\mathbb{Z})$ compute $P = U^t S U$, where $S \in \mathcal{S}_n^{>0}(\mathbb{Z}^n)$. Return $(vk = P, sk = U)$
Sign(sk, m) :	Compute $\mathbf{h} = H(m)$ compute $\mathbf{c} = U\mathbf{h}$ sample $\sigma' \leftarrow \mathcal{D}_{S, \rho/\sqrt{n}, c}$ compute $\sigma = U^{-1}\sigma'$. Return σ
Verify(vk, m, σ) :	Compute $\mathbf{h} = H(m)$ check that $\ \mathbf{h} - \sigma\ _P \leq \rho$ if so, return 1 else return 0

Fig. 4. Hawk Signature Scheme

As proved in [DvW22], the Hawk signature scheme is secure if the underlying $\text{ac} - \Delta\text{LIP}_s^{S, Q^{-1}}$ is hard. For the security proof, a class of quadratic forms is considered for which the signing is hard without the knowledge of the signing key. For example, for a given vector $\mathbf{h} \in \mathbb{R}^n/\mathbb{Z}^n$ there is no nearby vector. In this case, $\mathbf{h}$ is the full domain hash of the message m .

The following parameters were derived in [DvW22] for the Hawk signature scheme. Let $S \in \mathcal{S}_n^{>0}(\mathbb{Z})$ be a quadratic form. The signature scheme uses a sampling algorithm `DiscreteSample` that samples statistically close to $\mathcal{D}_{P, \rho/\sqrt{n}}(\mathbf{h} + \mathbb{Z}^n)$, where $\rho/\sqrt{n} \geq \nu_{2-\Theta(n)}([S])$ and any target $\mathbf{h} \in \mathbb{T}_q^n = \{0, \frac{1}{q}, \dots, \frac{q-1}{q}\}^n$. This algorithm exists according to Definition 3. The full-domain hash function is defined as follows: $H : \mathcal{M} \rightarrow \mathbb{T}_q^n$. The public parameters are

$$s \geq \max\{\lambda_n(S), \|B_S^*\| \cdot \sqrt{\ln(2n+4)/\pi}\}, \text{ and}$$

$$q := \left\lceil \frac{sn}{\rho} \cdot \sqrt{\ln(2n+4)/\pi} \right\rceil.$$

5.1 LIP-based Blind Signature Scheme

Definition 16 (LIP-based Blind Signature). *Our LIP-based blind signature scheme consists of the following three algorithms $\Sigma_{\text{LIP-BS}} = (\text{LIP-BS.KGen}, \text{LIP-BS.Sign}, \text{LIP-BS.Verify})$:*

- LIP-BS.KGen(1^λ): On input a security parameter 1^λ the algorithm samples a secret matrix $U \xleftarrow{\$} GL_n(\mathbb{Z})$ and computes the corresponding verification key $P = U^t S U$, where $S \in \mathcal{S}_n^{>0}(\mathbb{Z}^n)$. It outputs $\text{sk} = U$ and $\text{vk} = P$.
- LIP-BS.Sign(m, sk): This is an interactive protocol between a user and a signer. On input a message m , the user first blinds the message by first computing a hash value of it, i.e. $\mathbf{h} := H(m)$. Then she splits $\mathbf{h}$ into ℓ shares $\mathbf{H} = \{\mathbf{h}_1, \dots, \mathbf{h}_\ell\}$, s.t. there exists a subset $\{\mathbf{h}_{i_1}, \dots, \mathbf{h}_{i_\mu}\} \subset \mathbf{H}$ and integers $|a_i| \leq B$ (B and $\mu \ll \ell$ to be defined later) which satisfy $\mathbf{h} = \sum_{k=1}^{\mu} a_k \mathbf{h}_{i_k}$. Concretely, this can be implemented by uniformly sampling $\{i_1, \dots, i_\mu\} \subseteq [\ell]$, then $\mathbf{h}_{i_k} \xleftarrow{\$} \mathbb{T}_q^n$ and $a_k \xleftarrow{\$} [-B, B]$ for $k \in [\mu-1]$. Define $\mathbf{h}_{i_\mu} = \mathbf{h} - \sum_{k=1}^{\mu-1} a_k \mathbf{h}_{i_k}$ and $a_\mu = 1$. Finally sample the other $\mathbf{h}_i \xleftarrow{\$} \mathbb{T}_q^n$ for $i \in [\ell] \setminus \{i_1, \dots, i_\mu\}$. The user sends $\{\mathbf{h}_i\}_{i \in [\ell]}$ to the signer, while keeping the coefficients a_k and the selection $\{\mathbf{h}_{i_1}, \dots, \mathbf{h}_{i_\mu}\}$ secret. The signer uses the secret signing key U and computes $\mathbf{c}_i = U \mathbf{h}_i$ for all $i \in [\ell]$. Then she samples $\sigma'_i \leftarrow \mathcal{D}_{\mathbf{c}_i, \rho}$ and computes $\sigma_i = U^{-1} \sigma'_i$ for all $i \in [\ell]$. The signer sends these ℓ signatures to the user. After receiving the partial signatures, the user checks $\|\mathbf{h}_{i_k} - \sigma_{i_k}\|_P \leq \rho$ for $k = 1, \dots, \mu$ and reconstructs the signature on the unblinded message m as $\sigma = \sum_{k=1}^{\mu} a_k \sigma_{i_k}$. Finally, the user outputs the signature (σ, m) .
- LIP-BS.Verify(vk, σ, m): On input of a signature σ a verification key $\text{vk} = P$, the verifier runs the verification procedure of the underlying Hawk scheme, i.e. first computes $\mathbf{h} = H(m)$ and checks $\|\mathbf{h} - \sigma\|_P \leq \mu B \rho := \rho'$. If the norm condition is satisfied, the verifier outputs 1, otherwise outputs 0.

The security of this scheme is based on the CVP_γ (Definition 7). In particular, we need to select parameters ρ as in the Hawk scheme and $\rho' = \mu B \rho$ in such a way that

$$\mu B = \frac{\rho'}{\rho} \leq \gamma(n) ,$$

where the CVP_γ problem is hard. Blinding is assured by the fact that a small portion of the hashes is used. The total number of linear combinations

$$\sum_{k=1}^{\mu} a_k \mathbf{h}_{i_k}$$

with $|a_k| \leq B$ is

$$N = \binom{\ell}{\mu} (2B + 1)^\mu .$$

For instance, parameters $(\ell, \mu, B) = (130, 25, 1)$ will give a 128-bit N , and an acceptable security when using Hawk lattices for which the CVP_{25} problem is hard.

Theorem 3 (Correctness). *The blind signature scheme defined in 16 is correct.*

Signer $((U), P$	User P, m
	$h = H(m)$ for $i = 1, \dots, \ell$, $k = 1, \dots, \mu \ll \ell$ Choose $a_k \in \mathbb{Z}$, $ a_k \leq B$, $\mathbf{h}_i$
	$\xleftarrow{\mathbf{h}_1, \dots, \mathbf{h}_\ell}$ s.t. $\mathbf{h} = \sum_{k=1}^{\mu} a_k \mathbf{h}_{i_k}$
$\forall i \in [\ell]$ do: compute $\mathbf{c}_i = U \mathbf{h}_i$ Sample $\sigma'_i \leftarrow \mathcal{D}_{\rho, \mathbf{c}_i}$ Compute $\sigma_i = U^{-1} \sigma'_i$	
	$\xrightarrow{\sigma_1, \dots, \sigma_\ell}$
	Check $\ \mathbf{h}_{i_k} - \sigma_k\ _P \leq \rho \left(= \frac{\rho'}{\mu B} \right)$ Compute $\sigma = \sum_{k=1}^{\mu} a_k \sigma_{i_k}$ Output Hawk signature (σ, m)

Fig. 5. LIP-based Blind Signature (Fischlin framework)

Proof. For all $\mathbf{h}_{i_k}, \sigma_{i_k}$, $k \in [\mu]$ it holds that

$$\|\mathbf{h}_{i_k} - \sigma_{i_k}\|_P = \|U^{-1}(\mathbf{c}_{i_k} - \sigma'_{i_k})\|_S \leq \rho = \frac{\rho'}{\mu B} .$$

Taking a sum over all k yields:

$$\begin{aligned} \|\mathbf{h} - \sigma\|_P &= \left\| \sum_{k=1}^{\mu} a_k \mathbf{h}_{i_k} - \sum_{k=1}^{\mu} a_k \sigma_{i_k} \right\|_P \\ &\leq \sum_{k=1}^{\mu} \|a_k (\mathbf{h}_{i_k} - \sigma_{i_k})\|_P = \sum_{k=1}^{\mu} |a_k| \|\mathbf{h}_{i_k} - \sigma_{i_k}\|_P \leq \mu \cdot B \cdot \rho = \rho' . \end{aligned}$$

Theorem 4 (Unforgeability). Let $\Sigma_{\text{LIP-BS}} = (\text{LIP-BS.KGen}, \text{LIP-BS.Sign}, \text{LIP-BS.Verify})$ be the blind signature scheme in figure 5 instantiated with quadratic form $S \in \mathcal{S}_n^{>0}(\mathbb{Z})$, sampling parameter ρ' , public key parameter $s > 0$, and signature parameters ℓ, μ . Assume that $Q \in \mathcal{S}_n^{>0}(\mathbb{Z})$ has a dense rank sublattice as required by the Hawk signature sampler and let $\rho = \rho' / (\mu B)$.

If the problems $\text{ac-}\Delta\text{LIP}_s^{S, Q^{-1}}$ and CVP_{γ} are hard, then $\Sigma_{\text{LIP-BS}}$ are unforgeable in the random oracle model.

Proof. Let $\mathcal{A}$ be a probabilistic polynomial-time adversary against the one-more unforgeability of $\Sigma_{\text{LIP-BS}}$. We consider a sequence of games: **Game**₁, **Game**₂, **Game**₃. Let E_i denote the event that $\mathcal{A}$ wins in **Game** _{i} .

Game₁: This game is is the original one-more unforgeability experiment. The challenger samples $(U, P, S) \leftarrow \text{LIP-BS.KGen}(1^\lambda)$, and instantiates a random

oracle $H : \mathcal{M} \rightarrow \mathbb{T}_q^n$ and answers at most k signing queries made by $\mathcal{A}$. For a signing query on message m , let $h = H(M) \in \mathbb{T}_q^n$. The user writes

$$h = \sum_{r=1}^{\mu} a_r h_{i_r},$$

for some coefficients $a_r \in \mathbb{Z}$ with $|a_r| \leq B$ and some indices $i_r \in [\ell]$, where the vectors $h_1, \dots, h_\ell \in \mathbb{T}_q^n$ are sent to the signer. The signer computes for every $j \in [\ell]$:

$$c_j = U h_j, \quad \sigma'_j \leftarrow \mathcal{D}_{\rho, c_j}, \quad \sigma_j = U^{-1} \sigma'_j,$$

and returns $\sigma_1, \dots, \sigma_\ell$. The user checks $\|h_{i_r} - \sigma_{i_r}\|_P \leq \rho$ for every $r \in [\mu]$ and outputs

$$\sigma = \sum_{r=1}^{\mu} a_r \sigma_{i_r}.$$

Finally, $\mathcal{A}$ outputs $k+1$ message-signature pairs $(m_1, \sigma^{(1)}), \dots, (m_{k+1}, \sigma^{(k+1)})$. It wins if and only the messages are distinct, every pair passes the verification procedure and at most k signing interactions were initiated.

Game₂: This game is identical to **Game₁** except that the signing oracle is implemented without using the secret basis U . Whenever $\mathcal{A}$ initiates a signing interaction and sends $(h_1, \dots, h_\ell)$, the challenger answers by sampling, for each $j \in [\ell]$:

$$\tilde{\sigma}_j \leftarrow \mathcal{D}_{\rho, h_j}$$

and returns $\tilde{\sigma}_1, \dots, \tilde{\sigma}_\ell$. We claim that the adversary's view in both games is statistically indistinguishable. Indeed, in the real signing process, one has

$$\sigma'_j \leftarrow \mathcal{D}_{\rho, U h_j} \quad \text{and} \quad \sigma_j = U^{-1} \sigma'_j.$$

By the correctness of the Hawk-style sampler [DvW22], the distribution of σ_j is exactly the same as, or statistically indistinguishable from the distribution $\mathcal{D}_{\rho, h_j}$. Therefore, for every signing query and every adversarial view, the output distribution in **Game₂** matches that of the real signer. Therefore,

$$|\Pr[E_2] - \Pr[E_1]| \leq \text{negl}(\lambda).$$

Game₃: This is identical to **Game₂**, except that the public key is now sampled from the alternative distribution $P \leftarrow \mathcal{D}_s([Q^{-1}])$ instead of sampling from $\mathcal{D}_s([S])$. Since **Game₂** and **Game₃** use only public information and the signing oracle in **Game₂** is already simulated from the public centers h_j , any distinguisher between these two games yields a distinguisher for the assumption $\text{ac-}\Delta\text{LIP}_s^{S, Q^{-1}}$. Therefore:

$$|\Pr[E_3] - \Pr[E_2]| \leq \text{Adv}(\mathcal{A}_{\text{ac-}\Delta\text{LIP}_s^{S, Q^{-1}}}(\lambda)).$$

Here $\text{Adv}(\mathcal{A}_{\text{ac-}\Delta\text{LIP}_s^{S, Q^{-1}}}(\lambda))$ denotes the advantage of an adversary in breaking the $\text{ac-}\Delta\text{LIP}_s^{S, Q^{-1}}(\lambda)$ problem.

From a successful forgery to CVP_γ . Assume now that $\mathcal{A}$ wins in Game_2 with non-negligible probability. Then it outputs $k + 1$ valid signatures after at most k signing interactions. By the one-more condition, at least one valid forgery

$$(m^*, \sigma^*)$$

corresponds to a message m^* for which no signing interaction was completed. Since (m^*, σ^*) verifies, there exist coefficients $a_1^*, \dots, a_\mu^*$ with $|a_r^*| \leq B$ and vectors $h_{i_1}^*, \dots, h_{i_\mu}^*$ such that:

$$h^* = \sum_{r=1}^{\mu} a_r^* h_{i_r}^*$$

and

$$\sigma^* = \sum_{r=1}^{\mu} a_r^* \sigma_{i_r}^*, \quad \|h_{i_r}^* - \sigma_{i_r}^*\|_P \leq \rho \quad \text{for all } r \in [\mu].$$

Therefore

$$\|h^* - \sigma^*\|_P \leq \sum_{r=1}^{\mu} |a_r^*| \cdot \|h_{i_r}^* - \sigma_{i_r}^*\|_P \leq \mu B \rho = \rho'.$$

Therefore, a successful forgery provides a vector σ^* within distance ρ' of the target h^* in the P -norm.

Consequently, if ρ' is below the approximation threshold corresponding to γ , then from such a forgery one obtains a solution to CVP_γ for the lattice instance encoded by P . Hence

$$\Pr[E_2] \leq \text{Adv}(\mathcal{A}_{\text{CVP}_\gamma}(\lambda)).$$

Here $\text{Adv}(\mathcal{A}_{\text{CVP}_\gamma}(\lambda))$ denotes the advantage of an adversary of breaking the CVP_γ problem. □

The blindness of our signature scheme is based on a sparse bounded-coefficient variant of the Short Integer Solution (SIS) problem, defined below.

Definition 17 (Search Sparse bounded-coefficient SIS). Let n, q, ℓ, μ, B be parameters. The Search Sparse bounded-coefficient SIS problem over $\mathbb{T}_q^n$ is:

Given uniformly random vectors $\mathbf{h}_1, \dots, \mathbf{h}_\ell \in \mathbb{T}_q^n$ and a target $\mathbf{h} \in \mathbb{T}_q^n$, with bounded secret coefficients $a_1, \dots, a_\ell \in \mathbb{Z}$ such that

$$\sum_{i=1}^{\ell} a_i \mathbf{h}_i = \mathbf{h}$$

in $\mathbb{T}_q^n$, such that

$$|\{i \in [\ell] : a_i \neq 0\}| \leq \mu \quad \|\mathbf{a}\|_\infty \leq B,$$

for $\mathbf{a} = (a_1, \dots, a_\ell)$.

Relation to standard SIS. The search sparse bounded-coefficient SIS problem considered above can be viewed as an inhomogeneous variant of the standard SIS problem.

Let

$$A = (\mathbf{h}_1 \mid \cdots \mid \mathbf{h}_\ell) \in \mathbb{T}_q^{n \times \ell}$$

and let $\mathbf{h} \in \mathbb{T}_q^n$ be the target vector. The problem of finding coefficients $\mathbf{a} \in \mathbb{Z}^\ell$ such that

$$A\mathbf{a} = \mathbf{h} \quad \text{in } \mathbb{T}_q^n, \quad |\{i \in [\ell] : a_i \neq 0\}| \leq \mu, \quad \|\mathbf{a}\|_\infty \leq B,$$

is an instance of an *inhomogeneous* SIS problem.

This can be reduced to a homogeneous SIS instance by defining the modified matrix

$$A' = (A \mid -\mathbf{h}) \in \mathbb{T}_q^{n \times (\ell+1)}.$$

Then the above relation is equivalent to finding a vector

$$\mathbf{z} = \begin{pmatrix} \mathbf{a} \\ 1 \end{pmatrix} \in \mathbb{Z}^{\ell+1}$$

such that

$$A'\mathbf{z} = \mathbf{0} \quad \text{in } \mathbb{T}_q^n,$$

with the additional constraints

$$|\{i \in [\ell] : a_i \neq 0\}| \leq \mu, \quad \|\mathbf{a}\|_\infty \leq B, \quad z_{\ell+1} = 1.$$

Thus, solving the search sparse bounded-coefficient SIS problem yields a short, structured solution to a standard SIS instance. Conversely, the sparse variant can be seen as a restriction of SIS to solutions with bounded coefficients and small support.

In particular, under standard assumptions on the hardness of SIS, such structured solutions are believed to be hard to find for appropriate parameters [Kar72, FMRV22].

To prove blindness of our signature scheme, we introduce a decisional version of the sparse bounded-coefficient SIS problem.

Definition 18 (Decisional Sparse bounded-coefficient SIS). *Let n, q, ℓ, μ, B be parameters. The Sparse bounded-coefficient SIS problem over $\mathbb{T}_q^n$ is: given uniformly random vectors $\mathbf{h}_1, \dots, \mathbf{h}_\ell \in \mathbb{T}_q^n$ and a target $\mathbf{h} \in \mathbb{T}_q^n$, decide whether*

1. $\mathbf{h}$ is structured (i.e. as defined in the sparse bounded-coefficient SIS problem),
or
2. $\mathbf{h}$ is random over $\mathbb{T}_q^n$.

It is clear that an algorithm solving the search version can also solve the decisional version. Therefore, the decisional problem reduces to the search problem.

Theorem 5 (Computational blindness). *Assume that the decisional sparse bounded-coefficient SIS problem over $\mathbb{T}_q^n$ is hard for the parameter range (n, q, ℓ, μ, B) . Then the blind signature scheme*

$$\Sigma_{\text{LIP-BS}} = (\text{LIP-BS.KGen}, \text{LIP-BS.Sign}, \text{LIP-BS.Verify})$$

is blind in the random oracle model.

Proof. Suppose, by contradiction, that there exists a probabilistic polynomial-time adversary $\mathcal{A}$ that wins the blindness game with non-negligible advantage

$$\text{Adv}_{\Sigma_{\text{LIP-BS}}}^{\text{blind}}(\mathcal{A}) = \left| \Pr[b' = b] - \frac{1}{2} \right| \geq \varepsilon(\lambda)$$

for a non-negligible function ε .

We construct a probabilistic polynomial-time algorithm $\mathcal{B}$ that uses $\mathcal{A}$ to solve the decisional sparse bounded-coefficient SIS problem as given in Definition 18.

Algorithm $\mathcal{B}$ receives an instance

$$(\mathbf{h}, \mathbf{h}_1, \dots, \mathbf{h}_\ell)$$

of the sparse bounded-coefficient SIS problem and transmits it to $\mathcal{A}$ during the challenge signing interaction.

When $\mathcal{A}$ outputs two challenge messages (m_0, m_1) , algorithm $\mathcal{B}$ programs the random oracle so that

$$H(m_0) = \mathbf{h} \quad \text{and} \quad H(m_1) = \mathbf{h}'$$

for a random $\mathbf{h}'$ it is unlikely that there is a linear combination with small a_i s, which would pass the verification. If there were such a solution, then it would break the unforgeability of the signature scheme. Consequently, with overwhelming probability, no such short linear combination exists for the independent $\mathbf{h}'$.

Suppose that the adversary $\mathcal{A}$ can distinguish whether the target hash corresponds to m_0 or m_1 with non-negligible advantage, i.e.

$$|\Pr[b' = b] - 1/2| \geq \epsilon(\lambda) .$$

Then $\mathcal{A}$ must obtain some information about whether the transmitted shares $\mathbf{h}_1, \dots, \mathbf{h}_\ell$ that are submitted to the signing oracle contain a sparse bounded-coefficient representation of the target value $\mathbf{h} = H(m_0)$.

Indeed, conditioned on the public key and the oracle responses, the only difference between the two challenge transcripts is that in one case ($b = 0$), the target hash satisfies a hidden relation of the form

$$\mathbf{h} = \sum_{k=1}^{\mu} a_k \mathbf{h}_{i_k}, \quad |a_k| \leq B,$$

and in the other case ($b = 1$), the target hash $\mathbf{h}'$ is sampled independently of the transmitted shares.

If $\mathcal{A}$ outputs that the transcript corresponds to m_0 (i.e. $b = 0$), then $\mathcal{B}$ outputs *structured*; otherwise it outputs *uniform* (i.e. if $b = 1$). Hence $\mathcal{B}$ distinguishes the two cases with the same non-negligible advantage as $\mathcal{A}$.

This contradicts the assumed hardness of the decisional sparse bounded-coefficient SIS problem. Therefore no such adversary $\mathcal{A}$ exists, and the scheme is blind.

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A Our Blind Signature Scheme in Abe-Okamoto Framework

In Figure 6 we provide a construction of our blind signature scheme. We note that for this scheme to be secure we require the secret key matrices $U_b \xleftarrow{\$} GL_n(\mathbb{Z})$ to be symmetric.

<pre> BS.KGen(1^n) 101 : $\delta \xleftarrow{\\$} \{0, 1\}$ 102 : for $b \in \{0, 1\}$ do 103 : $U_b \xleftarrow{\\$} GL_n(\mathbb{Z})$ 104 : for $c \in \{-1, 1\}$ do 105 : $Q_b^{(c)} \leftarrow U_b^{cT} \tilde{Q} U_b^c$ 106 : $Q_b \leftarrow (Q_b^{(1)}, Q_b^{(-1)})$ 107 : return $(pk = (Q_0, Q_1), sk = (\delta, U_\delta))$ BS.S₁(sk) 201 : parse $(\delta, U_\delta) \leftarrow sk$ 202 : $\tilde{Y}_\delta \xleftarrow{\\$} \mathcal{D}_s([Q])^\kappa$ with $\tilde{V} \xleftarrow{\\$} GL_n(\mathbb{Z})^\kappa$ s.t. 203 : for $c \in \{-1, 1\}$ do 204 : $\tilde{Y}_\delta^{(c)} \leftarrow \tilde{V}^{(cT)} \tilde{Q} \tilde{Y}_\delta^c$ 205 : $\tilde{Y}_\delta \leftarrow (\tilde{Y}_\delta^{(1)}, \tilde{Y}_\delta^{(-1)})$ 206 : $\tilde{c}_{1-\delta} \xleftarrow{\\$} \{-1, 1\}^\kappa$ 207 : $\tilde{Y}_{1-\delta} \xleftarrow{\\$} \mathcal{D}_s([Q_{1-\delta}])^\kappa$ with $\tilde{R} \xleftarrow{\\$} GL_n(\mathbb{Z})$ s.t. 208 : for $c \in \{-1, 1\}$ do 209 : $\tilde{Y}_{1-\delta}^{(c)} \leftarrow \tilde{R}_{1-\delta}^{cT} Q_{1-\delta}^{c\tilde{c}_{1-\delta}} \tilde{R}_{1-\delta}^c$ 210 : $\tilde{Y}_{1-\delta} \leftarrow (\tilde{Y}_{1-\delta}^{(1)}, \tilde{Y}_{1-\delta}^{(-1)})$ 211 : $state_S \leftarrow (\tilde{V}_\delta, \tilde{c}_{1-\delta}, \tilde{R}_{1-\delta})$ 212 : return $(state_S, \rho_{S,1} = (\tilde{Y}_0, \tilde{Y}_1))$ BS.U₁(pk, msg, $\rho_{S,1}$) 301 : parse $(\tilde{Y}_0, \tilde{Y}_1) \leftarrow \rho_{S,1}$ 302 : for $b \in \{0, 1\}$ do 303 : $d_b \leftarrow \{-1, 1\}^\kappa$ 304 : parse $\tilde{Y}_b = (\tilde{Y}_b^{(1)}, \tilde{Y}_b^{(-1)})$ 305 : $Y_b \xleftarrow{\\$} \mathcal{D}_s([Q])^\kappa$ with $V_b \xleftarrow{\\$} GL_n(\mathbb{Z})^\kappa$ 306 : $Y_b \leftarrow V_b^{(T)} \tilde{Y}_b^{(d_b)} V_b$ 307 : $c \leftarrow H(Y_0 Y_1 msg)$ 308 : $\tilde{c} \leftarrow c \odot d_0 \odot d_1 \in \{-1, 1\}^\kappa$ 309 : $state_U \leftarrow (d_b, V_b)_{b \in \{0,1\}}$ 310 : return $(state_U, \rho_U = \tilde{c})$ </pre>	<pre> BS.S₂(state_S, ρ_U) 401 : parse $(\tilde{V}_\delta, \tilde{c}_{1-\delta}, \tilde{R}_{1-\delta}) \leftarrow state_S$ 402 : parse $\tilde{c} \leftarrow \rho_U$ 403 : $\tilde{c}_\delta \leftarrow \tilde{c} \odot \tilde{c}_{1-\delta} \in \{-1, 1\}^\kappa$ 404 : $\tilde{R}_\delta \leftarrow \tilde{V} \odot U_\delta^{(-\tilde{c}_\delta)}$ 405 : return $\rho_{S,2} = (\tilde{c}_b, \tilde{R}_b)_{b \in \{0,1\}}$ BS.U₂(state_U, $\rho_{S,2}$) 501 : parse $(d_b, V_b)_{b \in \{0,1\}} \leftarrow state_U$ 502 : parse $(\tilde{c}_b, \tilde{R}_b)_{b \in \{0,1\}} \leftarrow \rho_{S,2}$ 503 : for $b \in \{0, 1\}$ do 504 : $(c_b, R_b) \leftarrow (\tilde{c}_b \odot d_b, V_b \odot \tilde{R}_b^{(d_b)})$ 505 : $c' \leftarrow H((R_0^t Q_0^{S_0} R_0 R_1^t Q_1^{S_1} R_1 msg)$ 506 : if $c_0 \odot c_1 = c'$ 507 : return $\sigma = (c_b, R_b)_{b \in \{0,1\}}$ 508 : return $\sigma = \perp$ BS.Verify(pk, msg, σ) 601 : parse $(c_b, R_b)_{b \in \{0,1\}} \leftarrow \sigma$ 602 : $c' \leftarrow H((R_0^t Q_0^{S_0} R_0 R_1^t Q_1^{S_1} R_1 msg)$ 603 : if $c_0 \odot c_1 = c'$ 604 : return 1 605 : return 0 </pre>
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Fig. 6. Our blind signature scheme. We assume that the algorithms return $\perp$ and terminate if **parse** is not in the correct format.